



Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 1777
CALIBRATION DATE: 11-Feb-26

SBE 39 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

a0 = 7.597555e-05
a1 = 2.717482e-04
a2 = -2.238483e-06
a3 = 1.474992e-07

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
-1.5000	685042.4	-1.5000	0.0000
1.0000	610421.5	1.0000	-0.0000
4.5000	520936.2	4.5000	-0.0000
8.0000	446058.0	8.0000	-0.0000
11.5000	383184.7	11.5000	0.0000
15.0000	330214.0	15.0001	0.0001
18.5000	285439.9	18.5000	0.0000
22.0000	247472.4	22.0000	-0.0000
25.5000	215176.9	25.5000	-0.0000
29.0000	187623.0	28.9999	-0.0001
32.5000	164044.5	32.5001	0.0001

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1/\{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature

